# KEFFI AI – A MENTAL HEALTH CHATBOT


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## 1. ABSTRACT


Mental healthcare accessibility is a major global issue due to high therapy costs, long psychiatrist waitlists, and social stigma. Standard therapy offers only 1 hour per week, leaving patients unmonitored during the remaining 167 hours of weekly vulnerability. Keffi AI is a clinical digital therapeutics platform built to bridge this 167-hour gap by offering 24/7 continuous affective monitoring, hands-free voice interaction, and structured psychological interventions.


## 2. HIGHLIGHTS OF THE PROJECT

- **Fine-Tuned BERT 96-Emotion Model:** Identifies 96 distinct emotional categories from user text with 94.2% top-3 accuracy.
- **3-Tier Clinical Response Framework:** Combines Rogerian empathetic validation, biological explanations (amygdala and cortisol), and CBT grounding skills.
- **Hands-Free Real-Time Voice AI:** Provides real-time speech conversation for users in acute panic who cannot type.
- **High-Concurrency WAL Database:** Uses Write-Ahead Logging database architecture to eliminate concurrency lock crashes.
- **Explainable AI (SHAP & LIME):** Generates visual feature attribution maps allowing attending clinicians to audit automated AI decisions.
- **Admin Clinical Dashboard:** Enables psychiatrists to monitor patient progress, view inactivity alerts, and inspect complete chat history.

## 3. METHODOLOGY

Keffi AI uses a multimodal affective computing pipeline. When a user sends a text or speaks, input passes through a dual-model processing cascade. A fine-tuned BERT transformer classifies fine-grained emotion vectors, while an audio prosody analyzer measures pitch (F0), speech pace, and pauses. Conversation context is stored in a WAL relational database and vector memory tagged by patient ID. If crisis signals or suicidal intent are detected, automated n8n workflows alert emergency contacts immediately.

## 4. STEP BY STEP PROCEDURE OR ALGORITHM

**Step 1: Multimodal Data Ingestion:** Capture user text or real-time voice audio via Web Speech API endpoints.

**Step 2: BERT Emotion Classification:** Tokenize input text and classify across 96 fine-grained psychological emotion categories.

**Step 3: Isolated Context Retrieval:** Retrieve past conversation context from the WAL database using the patient ID.

**Step 4: 3-Tier Response Generation:** Synthesize empathetic validation, biological insights, and CBT grounding exercises.

**Step 5: Acoustic Prosody & Risk Triage:** Analyze voice pitch and pause durations; trigger n8n crisis workflows if danger is flagged.

**Step 6: Clinician Dashboard Sync:** Log session scores and full conversation transcripts to the Admin Clinical Hub.

## 5. DATASET USED IF ANY

- **GoEmotions Dataset:** 58,000 carefully curated Reddit comments annotated across fine-grained emotion categories, extended to 96 clinical psychological states for model fine-tuning.
- **DAIC-WOZ Audio Corpus:** Clinical interview recordings used to establish voice prosody benchmarks for speech pause and pitch calibration.

## 6. INPUT AND OUTPUT

**User Input:**

Spoken voice or written statement:

*"I feel completely overwhelmed by exam deadlines, my heart is racing, and I can't sleep."*

**System Output:**

- **Tier 1 Validation:** *"I hear how much pressure you're under. It is valid to feel overwhelmed."*
- **Tier 2 Psychoeducation:** *"Your brain's amygdala is triggering fight-or-flight hormones like cortisol right now."*
- **Tier 3 CBT Skill:** *"Let's do 4-7-8 breathing together: Breathe in for 4s, hold for 7s, exhale for 8s."*
- **Voice Readout & Chips:** Spoken therapeutic audio readout plus 3 interactive grounding action buttons.

## 7. CONCLUSION

Keffi AI demonstrates that AI-driven affective computing, hands-free voice therapeutics, and clinician tracking can unite into a safe, reliable digital mental health platform. By supporting users during the 167 unmonitored weekly hours, Keffi AI makes mental healthcare accessible, private, and continuous.

## DEVELOPED LIVE SYSTEM INTERFACE